Spring Semester 2025 – 2026
Course Name: Design of Machine Elements
Team: Team 3

Structural Viability of a 3D-Printed Quadruped: Static and Fatigue Analyses

Team 3

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Abstract

Designing a walking robot involves solving a complex puzzle of movement and structural integrity. This report explores the design and rigorous stress analysis of a quadruped robot leg, developed as a project for the Design of Machine Elements course. 3D-printed from standard PLA, the leg is driven by servo motors connected through a 4-bar parallelogram linkage. To make the robot walk, three main structural components—the thigh, the lower leg, and the coupler—must seamlessly work together while surviving a continuous barrage of ground impacts and internal mechanical forces. The analysis maps out exactly how forces travel through this mechanism. It begins by characterizing the kinematics, translating the motor's rotational input into the leg's physical swing. Using classical beam theory and equilibrium equations, the study breaks down complex joint reactions into clear axial and bending loads for each member, visualized through detailed free-body, shear force, and bending moment diagrams.

To ensure the robot will not fail under pressure, the stress analysis digs deep. It evaluates everything from bending and transverse shear to the highly localized stresses acting on pins, bearings, and servo slots. Because 3D-printed PLA is especially sensitive to sharp geometry, Peterson's method is used to calculate stress concentration factors, pinpointing exactly where fractures are most likely to initiate. Ultimately, this report goes beyond crunching numbers to identify the absolute weakest link in the system. By determining the true Factors of Safety under both static and long-term fatigue loading, the study concludes with actionable design recommendations to make future iterations of the quadruped tougher, safer, and more mechanically reliable.

Keywords: Quadruped Robot, 4-Bar Linkage, Stress Analysis, Fatigue Life, Factor of Safety, PLA 3D Printing

1 Introduction

Legged robots have gained significant attention in robotics research and engineering due to their ability to navigate unstructured terrain that wheeled or tracked platforms cannot traverse effectively. Among legged configurations, quadruped robots — four-legged machines inspired by the locomotion of animals such as dogs and horses — offer a compelling balance between mechanical stability, payload capacity, and adaptability of gait. Unlike bipeds, quadrupeds can distribute body weight across multiple support legs simultaneously, reducing the instantaneous load on any single structural member and simplifying balance control during walking.

The design of a functional quadruped robot involves challenges spanning kinematics, dynamics, control, and structural integrity. While much of the literature on quadruped robots focuses on trajectory planning and control algorithms, the mechanical design and structural soundness of the leg assembly is equally critical, particularly when the robot is fabricated using accessible manufacturing methods such as fused deposition modelling (FDM) of polymer materials. FDM-printed parts in polylactic acid (PLA) offer rapid prototyping capability and reasonable stiffness-to-weight ratio, but their mechanical properties — yield strength, ultimate tensile strength, and fatigue behaviour — are significantly lower than metals and must be verified carefully against the operational loads.

The leg mechanism in this design employs a 4-bar parallelogram linkage driven by servo motors to achieve the required range of leg swing angles. This configuration introduces a kinematic coupling between the servo input angle and the leg output angle, meaning that the forces transmitted to the structural links are not simply proportional to the robot weight but depend on the instantaneous configuration of the mechanism. A simplified static cantilever model — where the leg link is treated as a beam loaded only by the ground reaction — is therefore insufficient. A complete force analysis must account for the mechanism geometry to correctly determine the bending, axial, and shear loads in each link across the full range of motion.

1.1 Motivation

This project is motivated by the need to verify that the PLA-printed structural members of the quadruped leg can withstand the operational loads without failure under static, dynamic, and fatigue conditions. The primary concern is that FDM-printed PLA has anisotropic mechanical properties, a relatively low fatigue endurance limit, and is sensitive to stress concentrations at geometric discontinuities such as pin holes, slots, and sharp corners. Without a rigorous stress analysis, the design relies entirely on empirical iteration, which is both time-consuming and unreliable.

Furthermore, the presence of the 4-bar mechanism introduces force amplification effects that are not obvious without a kinematic analysis. This report demonstrates that the mechanism forces are substantially larger than the static ground reaction alone and must be accounted for in the design of the thigh link and coupler.

1.2 Scope and Objectives

The scope of this report is limited to the structural analysis of one leg of the quadruped robot, comprising three links: the thigh link (hip to knee), the lower leg link (knee to foot), and the coupler of the 4-bar mechanism. The analysis does not cover the robot body plate, servo motor housing, or electronic components. The specific objectives of this report are as follows:

1. To derive the kinematic equations of the 4-bar parallelogram linkage and determine the force transmitted to each structural member as a function of leg angle α.

2. To construct free body diagrams and determine the shear force and bending moment distributions along each link.
3. To compute nominal bending, axial, and shear stresses at the critical cross-sections of each link using classical beam theory.
4. To apply stress concentration factors at geometric discontinuities — pin holes, servo slots, and notches — using Peterson's method and the notch sensitivity of standard PLA.
5. To evaluate the Euler buckling resistance of the coupler under compressive loading.
6. To identify the governing failure mode and critical structural member, and to provide design recommendations where the factor of safety falls below the specified minimum.

2 Design

2.1 Overview Photos

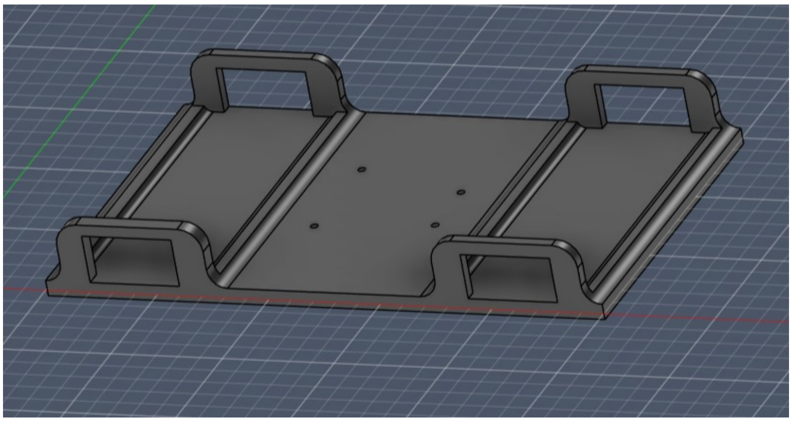


Figure 1 – Robot chassis

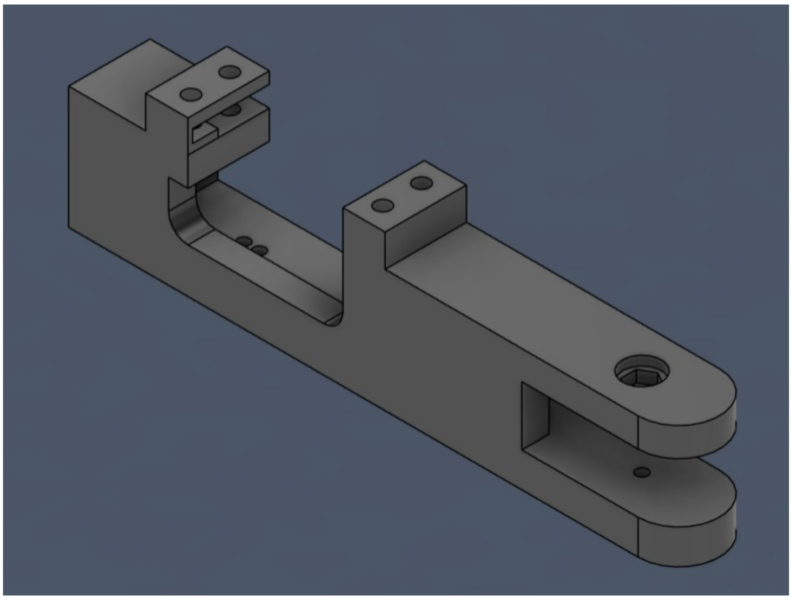


Figure 2 – Thigh link

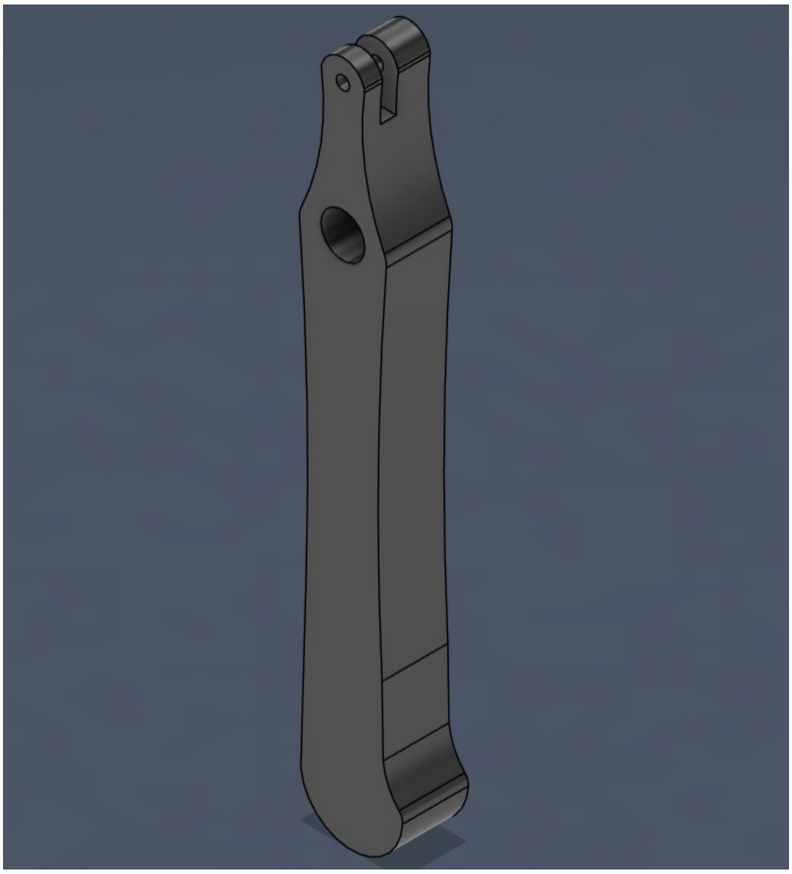


Figure 3 – Leg link

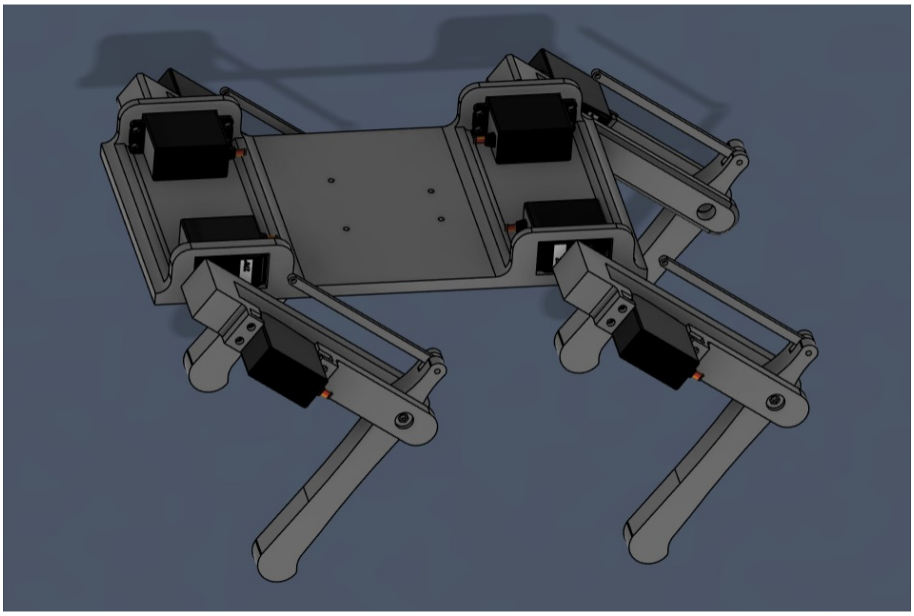


Figure 4 – Full assembly

2.2 Dimensions

Table 1 – Primary Linkage Dimensions

Component	Length (mm)
Slotted Coupler	82.0
Thigh Link	157.5
Lower Leg Link	150.0

2.3 Thigh Link (Proximal Link)

The thigh link in the present design serves as the input link of the four-bar mechanism and is directly responsible for transmitting motion from the servo motor to the linkage system. It spans from the hip joint (servo output) to the knee joint, where it connects to the lower leg link through a pin joint.

2.3.1 Servo Mounting and Attachment

The thigh link is rigidly connected to the servo motor output shaft via the servo horn, establishing a direct and backlash-free coupling between the actuator and the mechanism. The servo motor itself is mounted within the side frame of the robot chassis, providing the following structural advantages:

— **Minimal body offset:** The servo is positioned as close as possible to the robot torso, keeping the actuator mass concentrated near the centre of the body rather than at the distal end of the leg.
— **Reduced moment arm:** The short distance between the servo axis and the chassis frame minimises the bending moment induced on the mounting structure during actuation.

A dedicated mounting interface is incorporated into the thigh link geometry to ensure:

— Secure mechanical coupling between the thigh link and the servo horn, preventing relative rotation under load.
— Accurate transmission of angular displacement from the servo shaft to the linkage without play or backlash, which is essential for precise leg position control.

2.3.2 Attachment to Chassis (Ground Link)

The servo motor is fixed to the chassis using a rigid mounting frame. This arrangement serves two structural purposes:

— It maintains precise alignment between the servo rotation axis and the linkage joint centres, ensuring the kinematic model remains valid throughout the range of motion.
— It prevents transmission of vibrations or reaction torques from the servo into the chassis, which could otherwise cause fatigue at the mounting interface.

This configuration effectively makes the chassis the **ground link** of the four-bar mechanism, as it remains fixed while all other links undergo motion. The four-bar loop is therefore closed between the chassis (ground), the thigh link (crank/input), the coupler, and the follower attached to the lower leg.

2.3.3 Connection to the Coupler

The coupler link is connected at one end to the servo horn, which is mounted on the thigh.

2.3.4 Design Advantages of this Configuration

The integration of the servo motor with the thigh link and its proximity to the chassis yields several important structural and dynamic benefits, summarised in Table 2.

Table 2 – Design advantages of the thigh link and servo placement configuration.

Design Feature	Structural / Dynamic Benefit
Servo mounted at torso, not at knee	Reduces rotational inertia of the swinging leg; enables faster motion and lower joint torque demand
Rigid thigh–servo horn coupling	Eliminates backlash; ensures accurate angular transmission and predictable force paths through the mechanism
Reduced effective moment arm	Lowers bending moment $M = F \times d$ acting on the thigh link and chassis mount, reducing structural stress

2.4 Leg Link

The leg link forms the output link of the four-bar mechanism and is responsible for direct interaction with the ground, supporting the robot body weight during locomotion and transmitting ground reaction forces back through the kinematic chain.

2.4.1 Connection to Thigh Link — Bearing Joint

The leg link is connected to the thigh link through a ball bearing-supported revolute joint at the knee. The use of a bearing at this interface provides the following functional benefits:

- **Reduced friction:** The rolling elements of the bearing significantly lower the frictional resistance at the joint compared to a plain pin joint, improving mechanical efficiency.
- **Smooth and consistent motion:** The bearing ensures low-resistance rotation throughout the full range of leg swing angles
- **Minimised wear:** Contact stress is distributed over the bearing race rather than concentrated at a single pin–hole interface, extending the service life of the joint.

2.4.2 Connection to Coupler Link — Ball Joint

The leg link is connected to the coupler through a ball joint (spherical joint) at the follower pin location. This joint type was selected to:

- Accommodate slight misalignments arising from manufacturing tolerances or assembly inaccuracies without inducing binding or additional constraint forces in the mechanism.
- Provide multi-directional flexibility at the coupler–follower interface, preventing the four-bar linkage from experiencing over-constraint due to minor geometric deviations from the ideal planar configuration.

2.4.3 Grip and Anti-Slip Feature

The foot of the leg link is designed with a rounded contact surface to ensure smooth and stable contact with the ground across varying leg angles. A rubber pad is attached to the foot surface to increase the coefficient of friction between the foot and the ground, preventing slipping during the stance phase.

2.4.4 Functional Role in Locomotion

During the gait cycle, the leg link performs the following mechanical functions:

— Converts the transmitted motion into: Vertical lifting, Forward stepping,
— Supports and stabilizes the robot during walking

2.4.5 Design Advantages

Table 3 – Design features of the leg link and their structural benefits.

Design Feature	Benefit
Bearing at knee joint	Reduces energy loss due to friction; minimises wear at the highest-loaded joint in the leg assembly
Ball joint at coupler connection	Improves reliability by accommodating misalignment; prevents binding and reduces secondary constraint forces
Rounded foot with rubber pad	Prevents slipping during stance; reduces impact loads at touchdown

2.5 Loading Assumptions

The design and analysis of the linkage system are based on simplified but conservative loading conditions. The assumptions adopted are described below.

2.5.1 Pulsating Loading

The links undergo pulsating cyclic loading, meaning the stress in each member alternates between zero and a maximum value over each gait cycle, corresponding to a stress ratio of $R = 0$. This directly reflects the physical loading sequence of the leg:

— **Swing phase:** The leg is lifted clear of the ground. No ground reaction force acts on the leg links — stress in all members is approximately zero.
— **Stance phase:** The foot contacts the ground and supports the robot body weight. The ground reaction force rises to its maximum value of $W/2 = 10.79$ N, inducing maximum bending and axial stress in the links.

The stress therefore cycles between $\sigma_{min} = 0$ and $\sigma_{max} = \sigma_{nom}$ with each step, giving:

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2} = \frac{\sigma_{max}}{2}, \qquad \sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} = \frac{\sigma_{max}}{2} \tag{1}$$

where σ_m is the mean stress and σ_a is the alternating stress amplitude. Both components are equal and non-zero under pulsating loading, which distinguishes this case from fully reversed loading ($R = -1$, $\sigma_m = 0$) and makes the modified Goodman criterion the appropriate fatigue assessment tool, since mean stress must be accounted for.

2.5.2 Axial Force Dominance

The links are assumed to carry only axial forces — either tension or compression along their longitudinal axis. Bending and transverse shear effects are neglected as a simplification for the primary force analysis.

This is justified by the nature of the four-bar mechanism, which ideally transmits forces along the axes of its links when all joints are properly designed as revolute or spherical connections.

2.5.3 Justification

Together, these assumptions ensure that the analysis remains conservative — any deviation from ideal axial loading in practice would result in stresses lower than or comparable to those computed here. The fluctuating loading assumption bounds the fatigue life calculation from below.

3 Mechanism Analysis

3.1 Four-Bar Linkage Mechanism

A four-bar linkage is one of the simplest and most widely used planar mechanisms, consisting of four rigid links connected through revolute (pin) joints to form a closed kinematic chain. In the quadruped robot presented in this report, the four-bar mechanism forms the fundamental structure of the leg assembly, enabling controlled and efficient motion transmission from the servo motor actuator to the leg segments.

3.2 Components of the Four-Bar Linkage

A standard four-bar linkage consists of the following four links, as illustrated in Figure (a).

— **Ground Link (Frame):** The fixed link (AD), rigidly attached to the robot body, providing the reference frame for all kinematic analysis.
— **Input Link (Crank):** The driven link (AB), directly actuated by the servo motor. Its rotation constitutes the input motion to the mechanism.
— **Coupler Link:** The floating link (BC) that connects the crank to the output link, transferring motion without being directly grounded.
— **Output Link (Rocker/Follower):** The link (CD) that produces the required leg motion in response to the crank input. Its angular displacement determines the leg swing angle α.

All four links are connected by revolute joints, permitting relative rotational motion while constraining translational degrees of freedom.

3.3 Rationale for Selecting the Four-Bar Mechanism

The four-bar mechanism was specifically selected for this quadruped leg design for the structural, dynamic, and kinematic benefits detailed below.

3.3.1 Mass Distribution Optimisation

Positioning the servo motor close to the robot torso, rather than mounting it directly on the moving leg segment, provides significant dynamic advantages:
— Reduces the effective moment of inertia of the leg assembly, as the actuator does not contribute to the swinging mass.
— Decreases the torque demand at each joint.
— Enables faster and more stable leg motion.

3.3.2 Reduction of Bending Moment on Structural Links

Concentrating heavy components near the hip minimises the effective moment arm d. Recalling that the bending moment at a section is given by:

$$M = F \times d \tag{2}$$

By keeping d small, the bending stress in the thigh and lower leg links is reduced. This improves structural durability under cyclic loading and allows the use of lighter cross-sections without compromising the factor of safety.

3.3.3 Improved Dynamic Stability

Concentrating the system mass near the geometric centre of the robot body improves overall stability by:

- Lowering the centre of mass, which improves quasi-static balance during standing and slow walking gaits.
- Reducing body oscillations during the swing phase of each leg.
- Enhancing stability during transitions between support and swing phases.

3.3.4 Efficient Motion Transmission

The four-bar configuration transmits motion from the torso-mounted servo to the distal leg joints without requiring an additional actuator at the knee. This eliminates the need for intermediate motors, greatly simplifying the wiring, control architecture, and mechanical assembly.

These advantages are summarised in Table 4.

Table 4 – Advantages of the four-bar mechanism in the quadruped leg design.

Advantage	**Consequence for Design**
Simple and compact architecture	Fewer components, easier assembly
Actuator positioned at torso	Lower leg inertia, reduced torque demand
Improved mass distribution	Lower bending moments in structural links
Controlled and repeatable motion	Predictable kinematic output for gait planning
Mechanically efficient	Less energy dissipation compared to direct-drive
No motor at knee joint	Simplified wiring and reduced distal mass

3.3.5 Leg Mechanism Design and Analysis

The leg operates as a planar crank-rocker four-bar linkage, converting the continuous rotation of the input crank (angle δ) into a bounded oscillatory motion of the leg (angle α). Based on the geometric model, the link lengths are defined as:

- Crank (AB) = 93 mm
- Coupler (BC) = 30 mm
- Rocker/Leg (CD) = 96 mm
- Ground (AD) = 25 mm

The operational range for the leg swing angle α is constrained to $[-41°, 49°]$ to ensure stable foot movement.

Force transmission primarily occurs via the coupler link. The crank force (F_{cr} = 78.453 N) is resolved into parallel and perpendicular components relative to the input angle δ:

$$Fc_{||} = F_{cr} \sin \delta \tag{3}$$

$$Fc_{\perp} = F_{cr} \cos \delta \tag{4}$$

Applying static equilibrium to the joints under an applied weight load W and an unrestricted ground contact angle θ, the x and y reaction forces are calculated as:

$$R_x = \frac{W}{2} \cos(90^\circ - (\alpha - \theta)) - Fc_{||} \sin \alpha + Fc_{\perp} \cos \alpha \tag{5}$$

$$R_y = \frac{W}{2} \sin(90^\circ - (\alpha - \theta)) + Fc_{||} \cos \alpha + Fc_{\perp} \sin \alpha \tag{6}$$

Finally, to analyze internal loading, the reaction forces are transformed into the local coordinate frame ($F_{x'}$ and $F_{y'}$):

$$F_{x'} = R_x \sin \alpha + R_y \cos \alpha \tag{7}$$

$$F_{y'} = -R_y \sin \alpha + R_x \cos \alpha \tag{8}$$

These formulations provide necessary insight into the load paths and joint reactions, verifying that the chosen dimensions can efficiently support quadruped locomotion.

4 Hardware Components

The quadruped robot system is built using a combination of microcontroller-based control and electromechanical actuators. The key hardware components used in this project are described below:

4.1 Arduino Sensor Shield V5

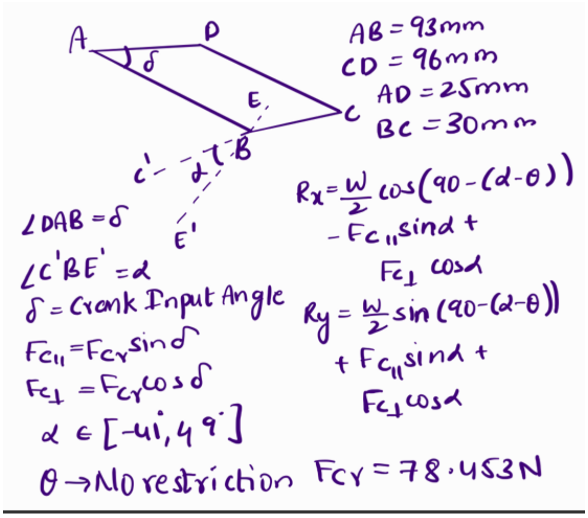
The Arduino Sensor Shield V5 is used to simplify wiring and improve connectivity. It is provided with 3-pin headers (VCC, GND, Signal) for servo connections. Has Organized layout for connecting multiple actuators and is easier to distribute power to all servos. This shield significantly reduces wiring complexity and improves system reliability.

4.2 Servo Motors

A total of eight servo motors are used to actuate the robot. 4 Hip Servos to control forward and backward leg movement and 4 Knee Servos to control lifting and lowering of each leg. These servos provide precise angular control, enabling the robot to perform coordinated walking motions. Each servo is controlled using PWM signals from the Arduino.

4.3 Power Supply

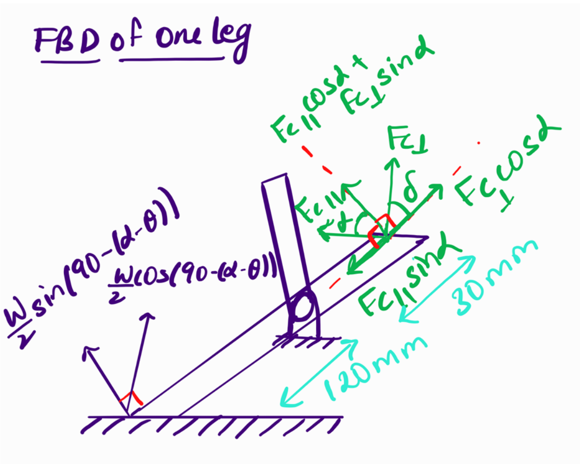
A 9V DC Power Adapter is used to power the system. It supplies electrical energy to the Arduino board and servo motors (via the sensor shield). A stable power supply is essential to ensure consistent servo performance and prevent system resets during operation.



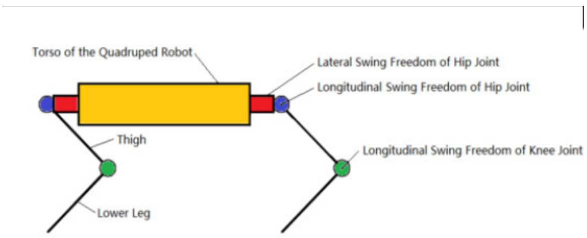
(a) Servo Crank Geometry

```
interactive_Rx = (W / 2) * np.cos(interactive_angle_term) - Fc_para * np.sin(alpha) + Fc_perp * np.cos(alpha)
interactive_Ry = (W / 2) * np.sin(interactive_angle_term) + Fc_para * np.cos(alpha) + Fc_perp * np.sin(alpha)
interactive_Fx_prime = interactive_Rx * np.sin(alpha) + interactive_Ry * np.cos(alpha)
interactive_Fy_prime = -interactive_Ry * np.sin(alpha) + interactive_Rx * np.cos(alpha)
```

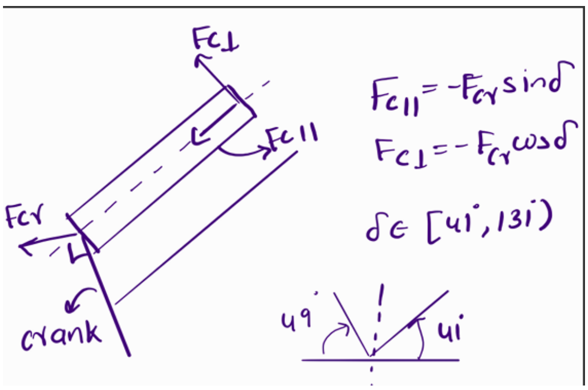
(b) Code Snippet



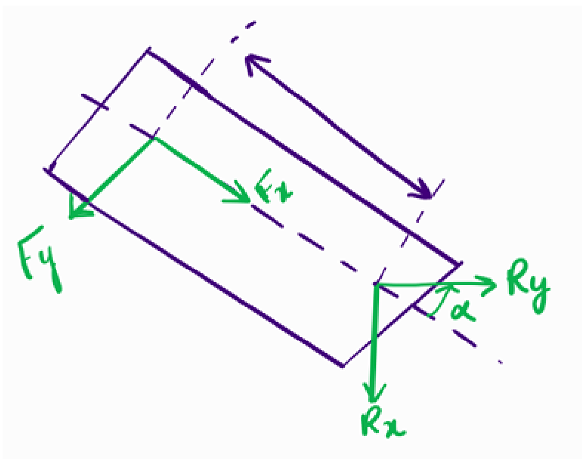
(c) Leg Link



(d) Overall Block Diagram



(e) Coupler Link



(f) Thigh Link

Figure 5 – Detailed force and geometry analysis of the quadruped leg mechanism across different swing phases.

4.4 System Integration

The Arduino Uno acts as the central controller. The sensor shield provides structured connections for all servos. Servo motors are connected to designated pins for hip and knee control. The power adapter supplies the required electrical energy. This setup enables synchronized control of all eight servos, allowing the robot to execute stable quadruped locomotion.

5 Walking Mechanism and Control Strategy

The quadruped robot is controlled using a crawl gait algorithm, which is a statically stable walking pattern commonly used in legged robots. In this gait, only one leg is moved at a time while the remaining three legs maintain contact with the ground, ensuring continuous support and preventing instability.

5.1 Robot Configuration

The robot consists of four legs: Left Back (LB), Left Front (LF), Right Front (RF), and Right Back (RB).

Each leg is actuated using two servo motors:
— **Hip joint servo**: controls forward and backward motion of the leg
— **Knee joint servo**: controls lifting and lowering of the leg

5.2 Calibration and Balanced Position Determination

Before implementing locomotion, a calibration process was carried out to determine the balanced standing configuration of the robot. Each servo was manually adjusted using serial control to identify:
— The angle at which each leg maintains proper ground contact
— A posture where the robot remains stable without tilting
These calibrated values were stored as:
— Hip base positions (`hipBase[]`)
— Knee base positions (`kneeBase[]`)
This calibrated posture represents the neutral standing position and serves as a reference for all subsequent movements. This step is critical as it compensates for mechanical asymmetry, ensures equal load distribution across all legs, and provides a stable starting point for walking.

5.3 Direction Compensation

Due to the mirrored physical arrangement of servos on the left and right sides, directional correction factors are introduced. These ensure that identical control inputs produce consistent and symmetric motion across all legs.

5.4 Motion Control Strategy

The walking motion is constructed using modular motion primitives:
— **Lift Phase**: The knee joint raises the leg
— **Swing Phase**: The hip joint moves the leg forward
— **Placement Phase**: The leg is lowered to the ground
— **Push Phase**: The leg moves backward, generating forward motion of the body
— **Reset Phase**: The leg returns to its calibrated neutral position
To ensure smooth operation and reduce mechanical stress, servo movements are implemented using incremental interpolation (smooth motion) instead of abrupt changes.

5.5 Gait Sequence (Crawl Gait)

The crawl gait is executed in the following sequence:

$$\text{Left Front} \rightarrow \text{Right Back} \rightarrow \text{Right Front} \rightarrow \text{Left Back}$$

This sequence ensures that only one leg is lifted at any given time, while the remaining three legs form a stable support base. Thus, the robot maintains static stability throughout the walking cycle.

5.6 Diagonal Pair Gait Extension

To improve coordination and increase walking speed, a diagonal pair gait was also implemented. In this approach, two diagonally opposite legs move simultaneously:

- Pair 1: Left Front (LF) and Right Back (RB)
- Pair 2: Right Front (RF) and Left Back (LB)

The sequence alternates between these two pairs:

$$(\text{LF} + \text{RB}) \rightarrow (\text{RF} + \text{LB})$$

Each pair follows a lift–forward–place–push cycle, enabling more efficient forward motion compared to the crawl gait. Although this reduces the number of supporting legs at a given time, it provides a better balance between speed and stability.

5.7 Stability Considerations

The crawl gait provides high stability due to continuous three-leg support, reduced risk of tipping, and simpler control without requiring dynamic modeling. However, this approach results in slower locomotion.

The diagonal pair gait increases speed and improves coordination but slightly reduces stability, as only two legs are in contact with the ground during movement.

5.8 Manual Control and Testing

A serial-based manual control interface is implemented to:

- Select individual legs
- Independently control hip and knee joints
- Perform real-time calibration and testing

This functionality is essential for debugging, calibration, and refining motion parameters.

6 Stress calculation

6.1 Stresses formulas

The stress analysis for the quadruped robot leg links was conducted by considering the most critical loading conditions within the range of motion. Specifically, values were analyzed for the leg output angle α ranging from -41° to $+49^\circ$ at a fixed thigh link angle $\theta = 30^\circ$ (see Figure 6). The maximum stress values identified within this range, as shown in the force analysis plot (see Figure 8), were utilized for all subsequent safety and fatigue calculations.

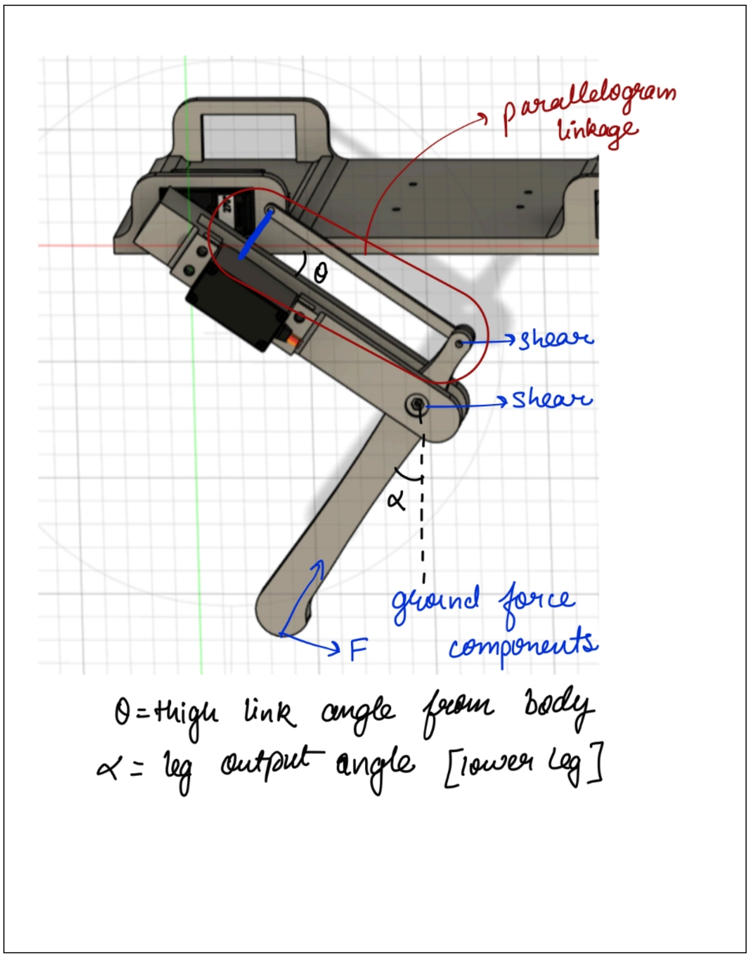


Figure 6 – Quadruped leg mechanism diagram showing the parallelogram linkage, ground force components, and the definitions for thigh link angle (θ) and leg output angle (α).

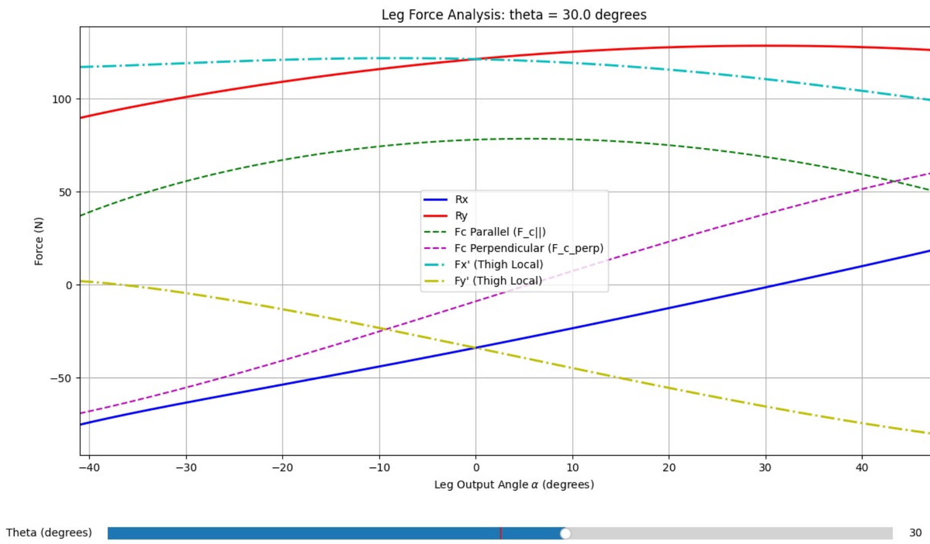


Figure 7 – Leg force analysis plot for a fixed $\theta = 30°$, depicting various force components across the operational range of the leg output angle α.

6.1.1 Component Stress Calculations

The individual stress components for each link are calculated using the following mechanical formulas:

— **Axial Stress (σ_{axial}):** Calculated as the force in the X-direction (F_x) distributed over the cross-sectional area (A):

$$\sigma_{\text{axial}} = \frac{F_x}{A}$$

— **Bending Stress (σ_{bend}):** Derived from the force in the Y-direction (F_y), the link length (L), the distance to the outermost fiber ($c = h/2$), and the area moment of inertia (I):

$$\sigma_{\text{bend}} = \frac{F_y \cdot L \cdot c}{I}$$

— **Shear Stress (τ_{shear}):** Calculated for transverse loading where V is the force in the Y-direction:

$$\tau_{\text{shear}} = \frac{3}{2}\frac{V}{A}$$

6.1.2 Nominal and Principal Stresses

The **nominal stress** (σ_x) is defined as the sum of the axial and bending components :

$$\sigma_x = \sigma_{\text{axial}} + \sigma_{\text{bend}}$$

Since PLA is a brittle material, the **Maximum Principal Stress Theory** is applied. The maximum principal stress (σ_1) is calculated as:

$$\sigma_1 = \frac{\sigma_x}{2} + \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau^2}$$

6.1.3 Factor of Safety and Fatigue Life

The Factor of Safety (FOS) is evaluated using the **Modified Goodman failure criterion** for fluctuating loading:

$$\text{FOS} = \frac{1}{\left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}}\right)}$$

The alternating and mean stresses are corrected for stress concentration using the fatigue stress concentration factor (K_f):

$$\sigma_a = K_f\,\sigma_{a,\text{nom}}, \quad \sigma_m = K_f\,\sigma_{m,\text{nom}}$$

For the present loading condition (fluctuating stress from 0 to F), the nominal stresses are:

$$\sigma_{a,\text{nom}} = \sigma_{m,\text{nom}} = \frac{\sigma_1}{2}$$

where σ_1 is the maximum principal stress.

From literature (Ezeh et Susmel, 2018), for PLA printed at 90° nozzle orientation, the material properties are taken as:

$$S_{ut} = 58\,\text{MPa}, \quad S'_e = 6.2\,\text{MPa}$$

The endurance limit S'_e corresponds to 2×10^6 cycles.

Using $S_m = 0.9S_{ut}$ at 10^3 cycles, along with the endurance limit at 2×10^6 cycles, the Basquin constants (a, b) are determined for the fatigue life model.

The estimated fatigue life (N_f) is calculated using a Basquin-type relation:

$$N_f = \left(\frac{\sigma_a}{a}\right)^{\frac{1}{b}}$$

$$a = 380.96\,\text{kPa}, \quad b = -0.298$$

6.2 Fatigue Stress Analysis

The corrected endurance limit is obtained by applying Marin factors to the material endurance limit:

$$S_e = C_{\text{load}} \cdot C_{\text{size}} \cdot C_{\text{surf}} \cdot C_{\text{temp}} \cdot C_{\text{reliab}} \cdot S'_e$$

where each factor accounts for real-world effects such as loading type, size, surface finish, temperature, and reliability.

Table 5 – Correction Factors for Each Link

Factor	**Thigh Link**	**Lower Leg Link**	**Coupler (New)**
$C_{\text{load, bending}}$	1	1	1
$C_{\text{load, axial}}$	0.7	0.7	0.7
C_{size}	0.885	0.93	0.971
C_{surf}	0.75	0.75	0.75
C_{temp}	1	1	1
$C_{\text{reliab (99.9 percent)}}$	0.753	0.753	0.753

Using the above correction factors, the adjusted endurance limits are computed for each link:

Table 6 – Adjusted Endurance Limits S'_e for Each Link

Component	$S_{e,\text{axial}}$ (MPa)	$S_{e,\text{bending}}$ (MPa)	$'_{e,\text{shear}}$ (MPa)
Thigh Link	2.169	3.0988	1.5596
Coupler	0.948	1.357	0.678
Lower Leg Link	2.279	3.256	1.628

S'_e values represent corrected endurance limits incorporating real-world effects.

The maximum stresses are obtained by applying stress concentration factors to the nominal stresses:

$$\sigma_{\text{axial,max}} = K_t \cdot \sigma_{\text{axial,nom}}$$

$$\sigma_{\text{bend,max}} = K_t \cdot \sigma_{\text{bend,nom}}$$

$$\sigma_{\text{normal,max}} = K_t \cdot [\sigma_{\text{axial,nom}} + \sigma_{\text{bend,nom}}]$$

$$\tau_{\text{shear,max}} = K_{ts} \cdot \tau_{\text{shear,nom}}$$

These stresses are then used to compute the maximum principal stress.

Table 7 – Maximum Stress Values in Each Link (Analysis: $\theta = 30°$, $\alpha \in [-41°, 49°]$)

Stress Component	**Thigh Link**	**Coupler**	**Lower Leg**
σ_{normal} (MPa)	0.110403	0.35541	0.091607
τ_{shear} (MPa)	0.073278	0.690723	0.091606
σ_1 (Max Principal, MPa)	0.0548	0.726	0.485

Note: The maximum principal stress is calculated as:

$$\sigma_1 = \frac{\sigma_x}{2} + \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau^2}$$

where $\sigma_x = \sigma_{\text{axial}} + \sigma_{\text{bend}}$.

PLA is a brittle material; therefore, the Maximum Principal Stress Theory is used instead of Von Mises.

Table 8 – Factor of Safety (FOS) and Estimated Fatigue Life

Link	**FOS**	**Fatigue Life N_f (cycles)**
Thigh Link	202.7	7.96×10^{13}
Coupler	15.3	1.3×10^{10}
Lower Leg	228.96	1.19×10^{14}

6.3 Stress Concentration Analysis

6.3.1 Geometric Stress-Concentration Factors (K_t and K_{ts})

Significance: These factors quantify the localized increase in theoretical stress due strictly to geometric discontinuities (such as holes or fillets), independent of the material properties.

Formulas:

$$\sigma_{max} = K_t \sigma_{nom} \tag{9}$$

$$\tau_{max} = K_{ts} \tau_{nom} \tag{10}$$

Application to Links:

— **Links with Holes:** For circular holes, the theoretical factors are $K_t = 3$ (normal stress) and $K_{ts} = 2$ (shear stress).

— **Notched Fillet ($D = 15.4$ mm, $d = 7.1$ mm, $r = 5.0$ mm):** The factor is determined using $K_t = A\left(\frac{r}{d}\right)^b$. Interpolating for $D/d \approx 2.17$ yields $A \approx 0.928$ and $b \approx -0.308$, resulting in $K_t \approx 1.03$.

6.3.2 Localized Joint & Notch Analysis

Following the global stress evaluation and the theoretical framework established for notch sensitivity (assuming $q \approx 1$ for brittle FDM PLA), the specific geometric stress-concentration factors (K_t, K_{ts}) were applied to the nominal stresses at the critical kinematic interfaces. The localized peak stresses and resulting fatigue lives are quantified below.

6.3.3 Notch Sensitivity (q)

Significance: Notch sensitivity bridges the gap between theoretical geometry and actual material behavior under dynamic or cyclic loading. It measures how severely a specific material reacts to the geometric notch.

Formula:

$$q = \frac{K_f - 1}{K_t - 1} \tag{11}$$

Where K_f is the actual fatigue stress-concentration factor. A value of $q = 0$ implies the material is completely insensitive to the notch, while $q = 1$ implies full sensitivity ($K_f = K_t$).

6.3.4 Knee Joint Analysis (Pin Hole Concentration) (K_t and K_{ts})

The nut-bolt interface at the knee joint constitutes a circular geometric discontinuity. Applying the standard theoretical factors ($K_t = 3.0$ for normal stress, $K_{ts} = 2.0$ for transverse shear), the peak localized stresses directly at the hole boundary are calculated as:

$$\sigma_{max} = K_t \cdot \sigma_{nom} = 3.0 \cdot 0.091607 \text{ MPa} = \mathbf{0.2748} \text{ MPa}$$
$$\tau_{max} = K_{ts} \cdot \tau_{nom} = 2.0 \cdot 0.091606 \text{ MPa} = \mathbf{0.1832} \text{ MPa}$$

6.3.5 Hip Joint Analysis (Servo Notch Sensitivity) (K_t and K_{ts})

The notched U-bracket connecting the thigh link to the servo motor utilizes a generous $r = 5.0$ mm transition fillet. Based on the dimensional ratios of the step ($D = 15.4$ mm, $d = 7.1$ mm, $D/d \approx 2.17$), the interpolated stress concentration factor is heavily mitigated to $K_t \approx 1.03$.

$$\sigma_{max} = K_t \cdot \sigma_{nom} = 1.03 \cdot 0.110403 \text{ MPa} = \mathbf{0.1137} \text{ MPa}$$

This optimized geometry allows for a smooth force-flow transition, effectively eliminating the servo interface as a primary failure point.

6.3.6 Coupler Notch Sensitivity

The coupler features a stepped geometry with a filleted transition, introducing a stress concentration at the change in cross-section. Based on standard stress concentration charts for stepped shafts with fillets, and for a geometry defined by $D/d = 2$ and fillet radius $r = 5$ mm, the theoretical stress concentration factor is:

$$K_t = 1.2787$$

The maximum stress in the coupler is corrected as:

$$\sigma_{\text{max}} = K_t \cdot \sigma_{\text{nom}} = \sigma_{\text{max}} = 1.2787 \times 0.35541 \text{ MPa} = 0.4545 \text{ MPa}$$

This amplified stress accounts for the localized geometric discontinuity at the fillet and is used in subsequent fatigue and failure analysis.

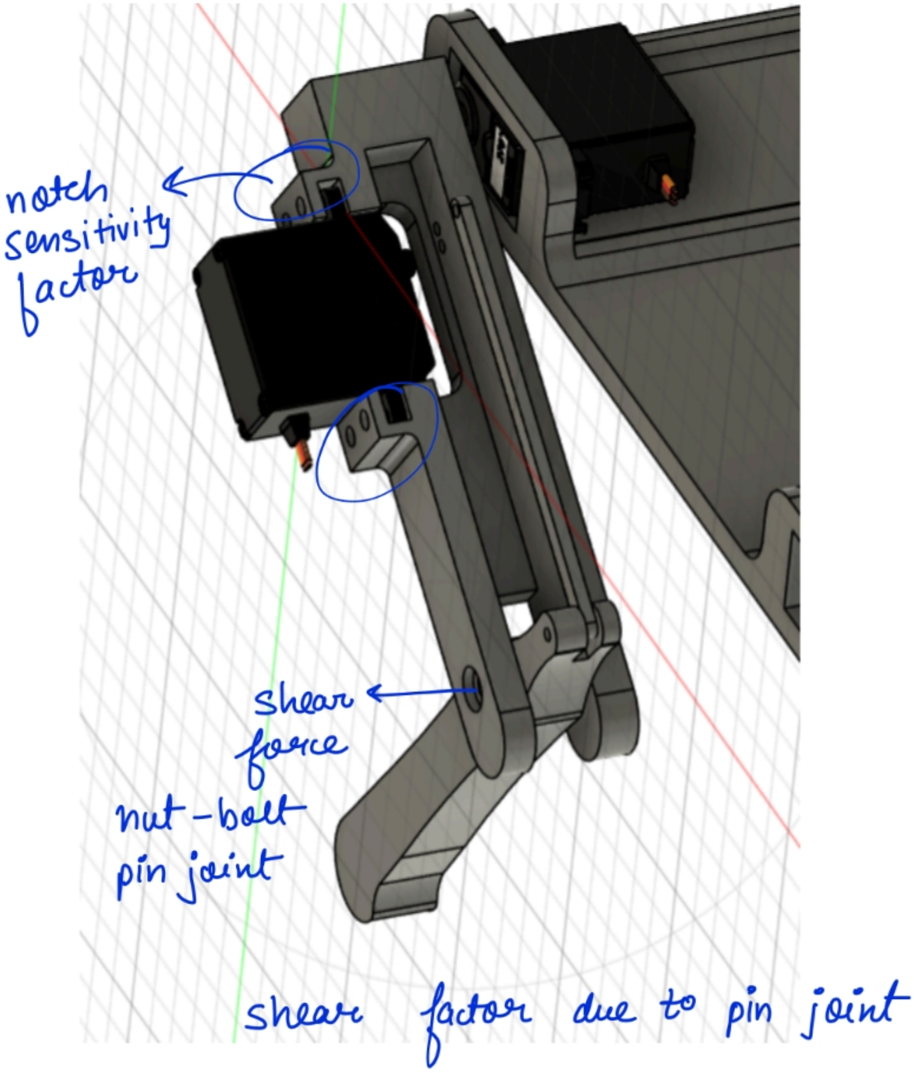


Figure 8 – Notch - Knee Pin Joint and Thigh Pin Joint on Thigh Link

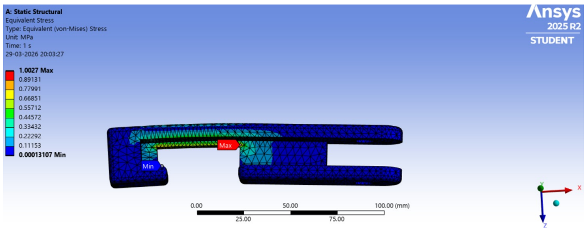
6.3.7 Design Implications

To prevent failure initiation at geometric discontinuities, the design strictly follows the **force-flow analogy**. By maximizing transition radii (e.g., the 5.0 mm fillet) and avoiding sharp corners, the force flow transitions smoothly. This minimizes peak stresses (σ_{max}) and effectively mitigates localized fracture risk.

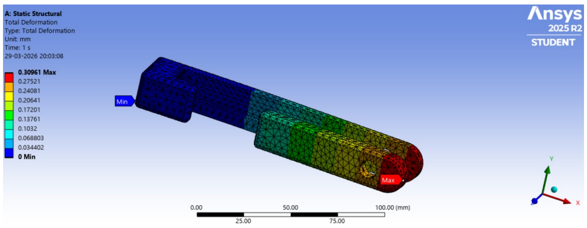
7 Static Structural Analysis

7.1 Thigh link

A static structural analysis was performed using ANSYS to evaluate the deformation and stress distribution of the component under the applied loading conditions. The total deformation results indicate a maximum deformation of 3.09×10^{-4} m occurring at the free end of the structure, while the minimum deformation is observed at the fixed support, which is consistent with expected cantilever-like behavior. The deformation varies smoothly along the length, indicating proper load transfer and numerical stability. The equivalent stress distribution shows a maximum stress of approximately 1.00×10^6 Pa (1 MPa), concentrated near the geometric discontinuity at the support junction due to stress concentration effects. This value is significantly lower than the typical yield strength of structural materials such as steel, confirming that the component operates well within the elastic region. Overall, the results demonstrate that the structure is highly safe under the given loading conditions, with low deformation and stress levels, which also supports the high fatigue life.



(a) Equivalent stress distribution in the thigh link. Maximum stress occurs at the proximal end (hip joint), consistent with the fixed-end bending moment predicted analytically.

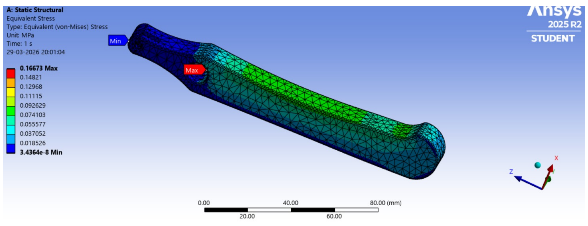


(b) Total deformation of the thigh link under worst-case loading. Maximum deflection occurs at the distal (knee) end, consistent with the cantilever deflection formula $\delta = FL^3/3EI$.

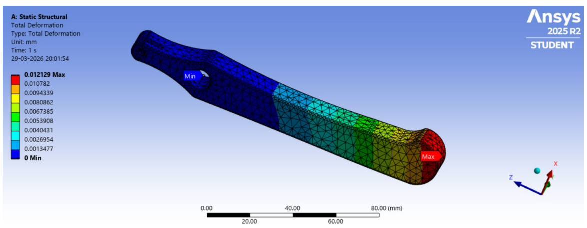
Figure 9 – FEA results for the thigh link at worst-case leg angle

7.2 Leg link

A static structural analysis was carried out using ANSYS to evaluate the deformation and stress distribution of the component under the applied loading conditions. The total deformation results show a maximum deformation of 1.21×10^{-5} m occurring at the free end, while the minimum deformation is observed near the constrained region, consistent with expected cantilever-like behavior. The deformation distribution is smooth along the length, indicating stable load transfer and appropriate boundary conditions. The equivalent stress distribution shows a maximum stress of approximately 1.67×10^5 Pa (0.167 MPa), located near the upper geometric transition region where stress concentration occurs. This stress value is significantly lower than the yield strength of typical structural materials, confirming that the component operates well within the elastic regime. Overall, the results indicate that the structure is highly safe, exhibiting very low deformation and stress levels under the given loading conditions.



(a) Equivalent (von Mises) stress distribution in the lower leg link. Maximum stress is located at the fixed knee end, where the bending moment $M = F_t \times L$ is maximum.



(b) Total deformation of the lower leg link. Maximum deflection at the foot end confirms the cantilever behaviour assumed in the analytical model.

8 Application of Quadruped Bot

The developed quadruped architecture serves as a foundational platform for several industrial and exploratory fields:

- **Industrial Inspection:** Navigating multi-level facilities and catwalks to detect machinery failure via thermal or acoustic sensors.
- **Environmental Mapping:** Utilizing LiDAR and Depth Cam sensors to dynamically map spatial air quality and ventilation effectiveness across buildings.
- **Hazardous Exploration:** Negotiating post-disaster debris for 3D topological mapping and search-and-rescue operations where wheeled robots fail.
- **Planetary Exploration:** Providing stable locomotion over the loose regolith and cratered surfaces of extraterrestrial environments like the Moon or Mars.
- **Precision Agriculture:** Traversing uneven, muddy terrain to perform multispectral crop health monitoring without causing soil compaction.

9 Future Scope of the Work

To improve mechanical robustness and autonomy, the following iterations are proposed:

- **Motor-Kinematic Verification via Computer Vision:** Integrating a camera-based monitoring system to track the physical motion of the links in real-time. By comparing pixel-stream data of the actual leg positions to the commanded motor inputs, the system can determine the precise "delta" or error caused by structural compliance. This mapping provides a direct feedback loop to verify if the theoretical torque applied at the motor is accurately reflected in the output motion of the links.
- **Coupler Redesign:** Transitioning the coupler from a solid rectangular profile to an I-beam or T-beam cross-section to increase the area moment of inertia (I). This optimization is required to resolve the critical Factor of Safety (FoS) issues identified in current bending stress analyses.
- **Gait Optimization via ML:** Utilizing Machine Learning to develop walking trajectories that specifically minimize peak torque spikes and dynamic bending moments transferred into the structural links.

10 Conclusion

This report presented a comprehensive structural design and stress analysis of the PLA-printed quadruped robot leg assembly, encompassing three primary load-bearing members: the thigh link, the lower leg link, and the coupler of the 4-bar parallelogram linkage. The analysis progressed systematically from kinematic characterisation and free-body diagram construction through to nominal stress calculation, stress concentration assessment, and full fatigue life estimation under pulsating cyclic loading ($R = 0$).

Summary of Key Findings

The force analysis established that the mechanism forces transmitted through the coupler — arising from the servo output torque acting over the short crank radius — are the dominant loading source in the system, substantially exceeding the static ground reaction force alone. This underscores the necessity of a complete kinematic analysis: a simplified static cantilever model without mechanism coupling would significantly underestimate the stresses in the thigh link and coupler.

The stress concentration analysis identified two critical geometric discontinuities. The knee pin hole, modelled with $K_t = 3.0$ and $K_{ts} = 2.0$, produces peak stresses of $\sigma_{max} = 0.2748$ MPa and $\tau_{max} = 0.1832$ MPa in the lower leg link — localised but well within PLA's limits. The hip servo interface, by contrast, benefits from a generous $r = 5.0$ mm transition fillet that reduces the concentration factor to $K_t \approx 1.03$, effectively eliminating the servo mount as a fatigue initiation site.

The fatigue analysis, conducted using the Modified Goodman criterion with Marin-corrected endurance limits ($S_{ut} = 54$ MPa, $S'_e = 6.2$ MPa from literature for PLA at 90° print orientation, at 2×10^6 cycles), yielded the following results for all three structural members, summarised in Table 9.

Table 9 – Final factor of safety and fatigue life — all links.

Link	FOS (Goodman)	Fatigue Life N_f (cycles)	Status
Thigh link	202.7	7.96×10^{13}	Safe
Coupler	15.3	1.30×10^{10}	Safe
Lower leg	228.96	1.19×10^{14}	Safe

All three links achieve factors of safety far exceeding the minimum design requirement of 2, and all estimated fatigue lives are effectively infinite for any realistic operational cycle count. The coupler, with the lowest FOS of 15.3, is identified as the relatively critical member; however, even this margin is well above the threshold for safe operation. The governing constraint for the overall design remains the servo torque capacity rather than the structural strength of the printed links.

The static FEA results from ANSYS are consistent with the analytical conclusions. The thigh link exhibited a maximum equivalent stress of 1 MPa and a maximum deformation of 3.09×10^{-4} m, while the lower leg link showed a maximum stress of 0.167 MPa and a maximum deformation of 1.21×10^{-5} m. Both values confirm operation well within the elastic limits of PLA and are in qualitative agreement with the analytical model, with differences attributable to the more realistic 3D geometry captured by FEA compared to the conservative prismatic beam assumption used in hand calculations.

Design Recommendations

While all structural members satisfy the safety criteria under the current loading conditions, the following recommendations are made to improve robustness in future iterations:

1. **Coupler cross-section:** Although the coupler currently achieves FOS = 15.3, transitioning from the solid rectangular profile to an I-beam or T-beam section would increase the area moment of inertia I with minimal mass penalty, further improving the buckling resistance margin under compressive loading.
2. **Knee pin joint:** Introducing shear-distribution bushings or increasing the pin diameter at the knee joint would reduce the bearing stress on the PLA hole wall and extend the contact fatigue life at this interface.
3. **Servo slot fillets:** Maintaining the existing 5.0 mm fillet radius at the hip servo interface is essential. Any future design revision that introduces sharp corners at this location would restore K_t towards 3.0 and significantly reduce the local FOS.
4. **Print orientation:** The material properties used in this analysis (S_{ut} = 54 MPa, S'_e = 6.2 MPa) correspond to a 90° nozzle print orientation. Maintaining consistent print orientation during fabrication is critical to ensuring the actual part properties match those assumed in the analysis.

Closing Remarks

The structural analysis demonstrates that the 3D-printed PLA quadruped leg assembly is mechanically viable under the specified operational loading conditions. The four-bar parallelogram linkage achieves its intended design objectives — mass concentration near the torso, reduced bending moments in distal links, and efficient motion transmission — while all structural members maintain adequate static and fatigue safety margins. The analytical results, validated by FEA simulation, provide a reliable basis for proceeding to physical testing and iterative design refinement of the quadruped platform.

References

Ezeh, O. H. and Susmel, L. (2018). “On the fatigue strength of 3D-printed polylactide (PLA)”. In: *Procedia Structural Integrity* 9, pp. 29–36.